

MECH5200 Numerical Methods for PDEs : *Video 6:* *Implementing 1D Finite Difference Equations*

David Willis and Xinfang Jin

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Thought Experiment

- Now that we have a governing equation $\rightarrow$ Finite Differences:

$$\frac{\partial^2 u}{\partial x^2} = f(x) \quad (1)$$

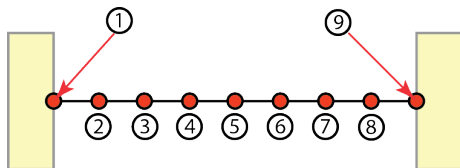
- With boundary conditions on both ends of:

$$u_L = u_R = 0 \quad (2)$$

- How do we solve this?

Step 1: Discretization – Setting up ODE/PDE

- Break string into a series of $N - 1$ equally sized "chunks", N nodes.



$$N = 9$$

$$\Delta x = 1/8$$

more generally

$$N = n$$

$$\Delta x = 1/(n-1)$$

Figure: Deriving the String Equation: First discretize and label the nodes. For now, spatial intervals/nodes must be equally spaced.

Step 2: Discretize the Governing Equation

- Governing equation: When this is satisfied everywhere, we have found the solution u :

$$\frac{\partial^2 u}{\partial x^2} = f$$

- We can write a discrete finite difference equation – this allows us to satisfy the equation approximately at each location/node where the finite difference equation is written:

$$\frac{u_{i-1} - 2u_i + u_{i+1}}{(\Delta x)^2} \simeq f_i$$

- Finite Differences: Determine the solution u that satisfies the approximate difference equations at the nodes of the discrete domain.

Step 2: Discretize the Governing Equation

- Let's write an equation for EACH of the nodes *inside* the domain:

$$\begin{aligned}\frac{u_1 - 2u_2 + u_3}{(\Delta x)^2} &\simeq f_2 \\ \frac{u_2 - 2u_3 + u_4}{(\Delta x)^2} &\simeq f_3 \\ \frac{u_3 - 2u_4 + u_5}{(\Delta x)^2} &\simeq f_4 \\ &\vdots \\ \frac{u_{N-2} - 2u_{N-1} + u_N}{(\Delta x)^2} &\simeq f_{N-1}\end{aligned}$$

Step 2: Discretize the Governing Equation

- Let's re-write the equations by factoring them as the summation of scaled column vectors:

$$\begin{bmatrix} 0 \\ \frac{1}{\Delta x^2} \\ 0 \\ 0 \\ 0 \\ \vdots \end{bmatrix} u_1 + \begin{bmatrix} 0 \\ \frac{-2}{\Delta x^2} \\ \frac{1}{\Delta x^2} \\ 0 \\ 0 \\ \vdots \end{bmatrix} u_2 + \begin{bmatrix} 0 \\ \frac{1}{\Delta x^2} \\ \frac{-2}{\Delta x^2} \\ \frac{1}{\Delta x^2} \\ 0 \\ \vdots \end{bmatrix} u_3 + \begin{bmatrix} 0 \\ 0 \\ \frac{1}{\Delta x^2} \\ \frac{-2}{\Delta x^2} \\ \frac{1}{\Delta x^2} \\ \vdots \end{bmatrix} u_4 + \dots = \begin{bmatrix} 0 \\ f_2 \\ f_3 \\ f_4 \\ f_5 \\ \vdots \end{bmatrix}$$

Step 3: Form a Linear System of Equations

- The scaled column vectors is a representation of matrix vector multiplication:

$$\begin{bmatrix} 0 & 0 & 0 & 0 & \dots \\ \frac{1}{(\Delta x)^2} & \frac{-2}{(\Delta x)^2} & \frac{1}{(\Delta x)^2} & 0 & \dots \\ 0 & \frac{1}{(\Delta x)^2} & \frac{-2}{(\Delta x)^2} & \frac{1}{(\Delta x)^2} & \dots \\ \vdots & \vdots & \vdots & \vdots & \vdots \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ \vdots \\ u_N \end{bmatrix} = \begin{bmatrix} f_1 \\ f_2 \\ f_3 \\ f_4 \\ \vdots \\ f_N \end{bmatrix}$$

- **Each column:** represents the connection of the column unknown between each FD equation containing that unknown.
 - Columns relate to individual unknowns.
- **Each row:** For each row, the finite difference equation for the second derivative “connects” 3-unknowns in the equation.
 - Rows represent each node’s FD equation.
 - The unknowns are linked to other nodes through the $i + 1$ and $i - 1$ nodes.

Step 3: Form a Linear System of Equations

- Recognizing that we have a linear system of equations, let's check it using a row-column multiplication:

$$\begin{pmatrix} \text{eqn Node 1} \\ \text{eqn Node 2} \\ \text{eqn Node 3} \\ \text{eqn Node 4} \\ \vdots \\ \text{eqn Node } N \end{pmatrix} = \frac{1}{(\Delta x)^2} \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & \dots \\ 1 & -2 & 1 & 0 & 0 & \dots \\ 0 & 1 & -2 & 1 & 0 & \dots \\ 0 & 0 & 1 & -2 & 1 & \dots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & 0 & \dots \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ \vdots \\ u_N \end{pmatrix} = \begin{pmatrix} f_1 \\ f_2 \\ f_3 \\ f_4 \\ \vdots \\ f_N \end{pmatrix}$$

- Expanding out the third row/equation of the linear system:

$$\text{eqn Node 3} = \frac{1 * u_2 - 2 * u_3 + 1 * u_4}{(\Delta x)^2} = f_3$$

- This checks out – the pattern we see in the matrix corresponds to the finite difference formula!!
- The only remaining problem is that this matrix has ALL zeros in rows 1 and N .

Step 4: Incorporate Boundary Conditions

- Introduce boundary conditions.
- What are the boundary conditions for this problem?
 - What do we know about u or du/dx at $x = 0$ or $x = 1$?

Step 4: Incorporate Boundary Conditions

- The boundary conditions: No deflection of the string at the two end-points:

- $u = 0$ at $x = 0$ and $x = 1$

- The first boundary condition, $u_1 = 0$ is:

$$1 * u_1 + 0 * u_2 + 0 * u_3 + \dots + 0 * u_N = 0 \quad (3)$$

- The second boundary condition, $u_N = 0$ is:

$$0 * u_1 + 0 * u_2 + 0 * u_3 + \dots + 1 * u_N = 0 \quad (4)$$

- To implement these B.C.'s in the A-matrix:

- A-matrix entry $A(1, 1) = 1$, and $f_1 = 0$
 - A-matrix entry $A(N, N) = 1$, and $f_N = 0$

Note: Because we factored out $\frac{1}{(\Delta x)^2}$, we will adjust accordingly:

$$\begin{pmatrix} \text{eqn Node 1} \\ \text{eqn Node 2} \\ \text{eqn Node 3} \\ \text{eqn Node 4} \\ \vdots \\ \text{eqn Node N} \end{pmatrix} = \frac{1}{(\Delta x)^2} \begin{pmatrix} (\Delta x)^2 & 0 & 0 & \dots & 0 \\ 1 & -2 & 1 & 0 & \dots & 0 \\ 0 & 1 & -2 & 1 & \dots & 0 \\ 0 & 0 & 1 & -2 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & \dots & (\Delta x)^2 \end{pmatrix} \begin{pmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ \vdots \\ u_N \end{pmatrix} = \begin{pmatrix} 0 \\ f_2 \\ f_3 \\ f_4 \\ \vdots \\ 0 \end{pmatrix}$$

Class Coding Activity

- Solve the linear system of equations using MATLAB.
- We will use the 'backslash' operator in MATLAB:

Solution

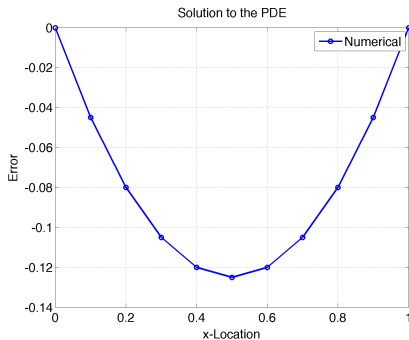


Figure: Solution for $f = 1$, $N = 11$

Solution

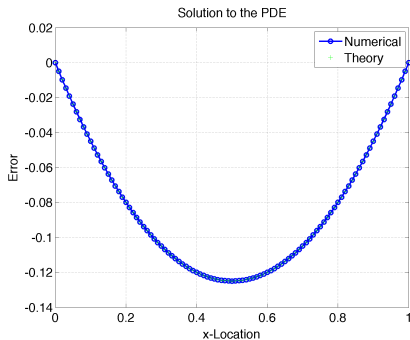


Figure: Solution for $f = 1$, $N = 101$

What have we learned?

- Finite difference expressions:
 - Approximate the "governing" PDE/ODE expression at chosen nodes in the domain
 - Determines the solution that satisfies the (FD approximated) equation at those nodes
- The general steps for a finite difference problem are:
 - Step 1: Discretize the domain (in 1D: $N - 1$ intervals, N nodes or points).
 - Step 2: Write finite difference equations that approximate governing equation at each internal node of the problem
 - Step 3: Form a linear system ($Ax = b$)
 - Step 4: Enforce the boundary conditions where relevant
 - Step 5: Solve the system of equations